Investigating Environmental Exposome Risk Factors of ALS using Double Machine Learning

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Introduction

- Ecological studies suggest that **environmental exposures** may contribute to geographic variation in ALS risk.
- Individual-level evidence linking specific toxic exposures to ALS remains limited.
- Exposure mixtures, nonlinear relationships, and high-dimensional confounding make conventional regression challenging.
- Linking **longitudinal residential history** with **EPA RSEI** exposure estimates enables patient-level characterization of the exposome.
- **Double Machine Learning (DML)** combines flexible ML-based confounding adjustment with interpretable estimates of exposure–ALS associations.

Objective: estimate the association between time-varying residential exposure to TRI-reported toxic chemicals, measured by RSEI scores, and ALS risk, using DML to adjust flexibly for confounding.

Exposure Data: EPA TRI and RSEI Score

The **Toxics Release Inventory (TRI)** records facility-level releases of more than 650 toxic chemicals to air, water and land. **RSEI** turns these releases into a toxicity-weighted score that accounts for emission amount, chemical toxicity and environmental fate.

EPA TRI (Toxics Release Inventory)

What it is

A national database of toxic chemicals released to air, water, land, and transferred off-site from industrial facilities.



Key features

- Facility-level reporting of > 650 toxic chemicals
- Annual reports on releases and waste management activities
- Includes air emissions, water discharges, land releases, and off-site transfers
- Publicly available data since 1987

TRI Data Example

- Air Emissions (e.g., pounds/year)
- Water Discharges (e.g., pounds/year)
- Off-site Transfers (e.g., pounds/year)

RSEI (Risk-Screening Environmental Indicator) Score

What it is

A cumulative, toxicity-weighted score that estimates the potential risk of chemical exposures to nearby communities from TRI-reported facility emissions.

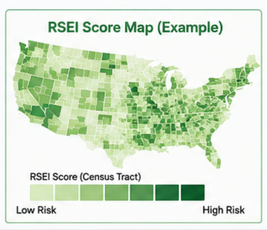


From emissions to potential community risk

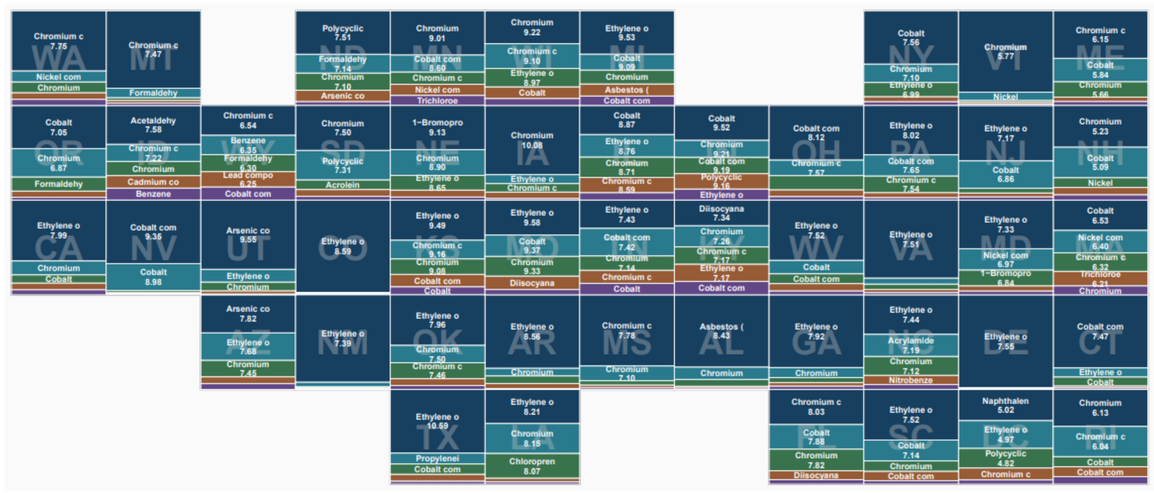
Key features

- Integrates TRI data with chemical toxicity information and environmental fate
- Calculated for U.S. census tracts
- Accounts for emissions amount, toxicity, and proximity (air, water, land pathways)
- Higher scores indicate greater potential risk to the surrounding population

RSEI Score Map (Example)



RSEI score summary by state. Chemicals with the highest RSEI scores in each state, shown as a tile map. Chromium compounds, cobalt and ethylene oxide rank at or near the top in most states.



Linking RSEI Scores with Patients

Patient addresses are geocoded by year to **census block groups (CBG)**. RSEI values on the 810 m × 810 m grid are interpolated to each CBG with **areal Tobler interpolation**, giving each patient a **time-varying exposure** that updates whenever they move.

Attributing TRI-reported Chemical Exposures to Patients' Living Neighborhood

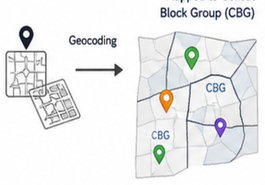
From addresses to time-varying, neighborhood-level exposure

1 Geocode patient addresses and map to census block group

Patient addresses (by year)

Patient ID	Address (Year)
P001	123 Main St (2010)
	456 Oak Ave (2014)
	789 Pine Rd (2018)
P002	321 Elm St (2012)
...	

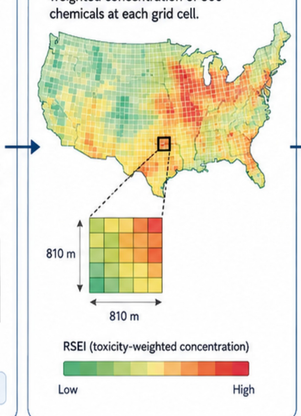
Mapped to Census Block Group (CBG)



Output: Patient-year → Census Block Group (CBG)

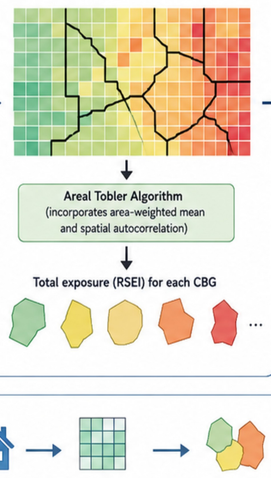
2 EPA RSEI raster database (810m × 810m grid)

EPA provides RSEI (Risk-Screening Environmental Indicators) – toxicity-weighted concentration of 500+ chemicals at each grid cell.



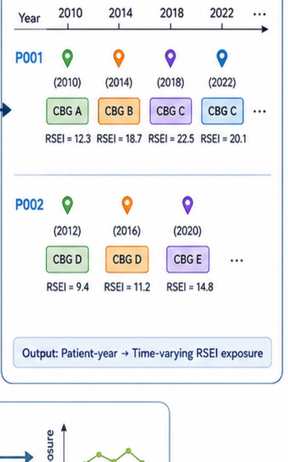
3 Areal Tobler interpolation to estimate exposure for each CBG

Overlay CBG polygons on the RSEI grid. Apply Areal Tobler algorithm to interpolate the total exposure for each CBG based on the overlapping grid cells and their spatial proximity.



4 Time-varying exposure assignment to patients

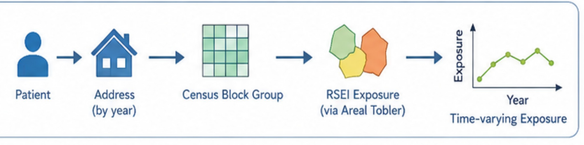
Link patient-year CBG to the corresponding CBG-level exposure.
If a patient moves, exposure updates accordingly.



Output: Patient-year → Time-varying RSEI exposure

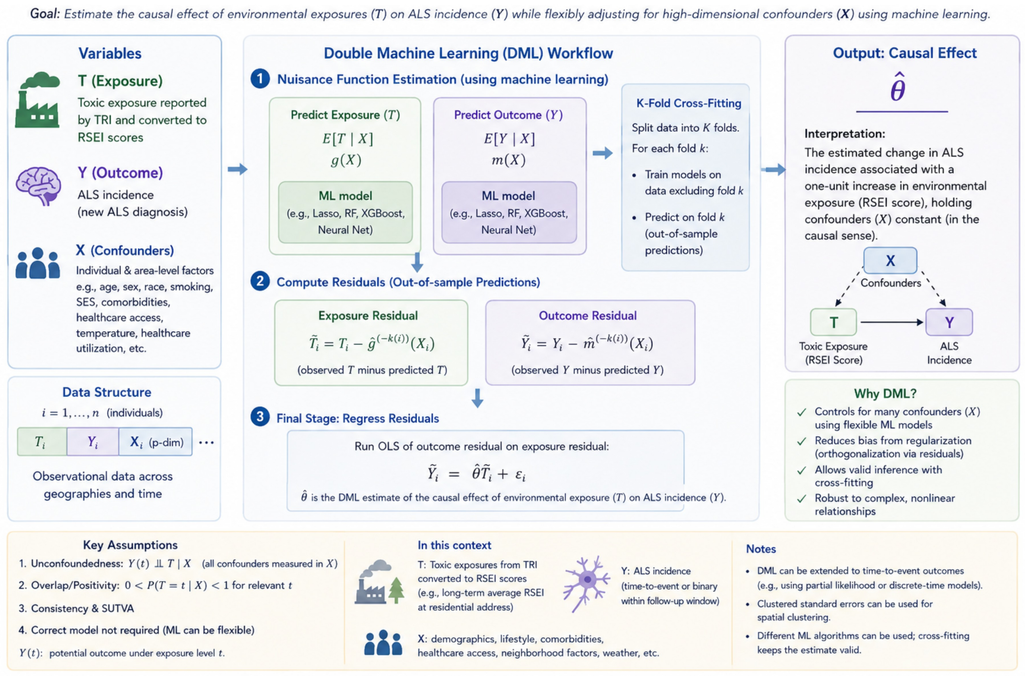
Final Result

Each patient has a time-varying, TRI-derived exposure estimate based on their annual living location (census block group).



Double Machine Learning

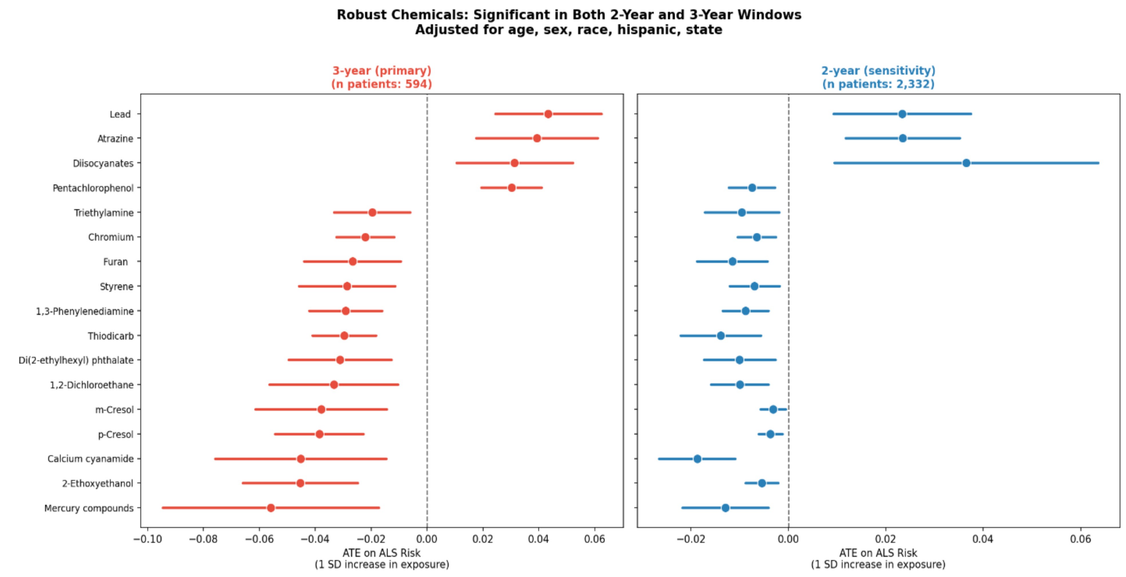
*Goal: Estimate the causal effect of environmental exposures (**T**) on ALS incidence (**Y**) while flexibly adjusting for high-dimensional confounders (**X**) using machine learning.*



Statistical Analysis

- **Exposure assessment:** chemical-specific RSEI exposures were aggregated over pre-index windows of 2 to 5 years and standardized to allow comparison across chemicals.
- **Confounder adjustment:** age, sex, race, ethnicity, dual Medicare–Medicaid eligibility, rural–urban commuting area (RUCA), Area Deprivation Index (ADI) and calendar year.
- **Cross-fitting:** five-fold cross-fitting with three repetitions to reduce overfitting and regularization bias.
- **Spatially robust inference:** geographic grouping was built into cross-fitting, with county-clustered robust standard errors derived from orthogonal scores to account for within-county dependence.
- **Quality control:** chemical-specific exposure sparsity, variability and nuisance-model performance were assessed to flag unreliable estimates.
- **Multiple testing:** Benjamini–Hochberg false discovery rate (FDR) correction across chemical comparisons.
- **Sensitivity analyses:** associations were evaluated across 2–5-year exposure windows, including common-cohort comparisons; chemicals with consistent positive associations across windows were prioritized for further investigation.

Results



1.02–1.06×
ALS risk per 1 SD increase in RSEI score
3-year window (primary)

1.01–1.07×
ALS risk per 1 SD increase in RSEI score
2-year window (sensitivity)

Positive in both windows

Lead
Consistent with prior evidence

Atrazine
Signal warrants further study

Diisocyanates
Signal warrants further study

Discussion and Conclusion

- **Exposure misclassification:** incomplete residential history limits characterization of cumulative and long-latency exposures.
- **Area-based exposure:** an improvement over ecological studies, but RSEI still modeled neighborhood-level exposure, not personal dose.
- **Novel signals need validation:** prior evidence supports lead (Fang et al., 2010; Meng et al., 2020); the atrazine and diisocyanate associations are hypothesis-generating.
- **Residence is only part of the exposome:** occupational exposure may matter more for some industrial chemicals.
- **Residual confounding** remains possible despite DML, especially from occupation and other unmeasured exposures.

Linking longitudinal residential histories with EPA RSEI data improves patient-level characterization of the ALS exposome. DML enables rigorous, flexible, high-dimensional-confounder-adjusted screening of environmental risk factors, providing a scalable framework for longitudinal exposome research.

References

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2. U.S. Environmental Protection Agency. Risk-Screening Environmental Indicators (RSEI) Model.
3. Fang F, et al. Association between blood lead and the risk of amyotrophic lateral sclerosis. *Am J Epidemiol.* 2010;171(10):1126–1133.
4. Meng E, et al. Population-based study of environmental/occupational lead exposure and amyotrophic lateral sclerosis: a systematic review and meta-analysis. *Neurol Sci.* 2020;41(1):35–40.

Acknowledgments

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